EE115 Analog Circuits
模拟电路

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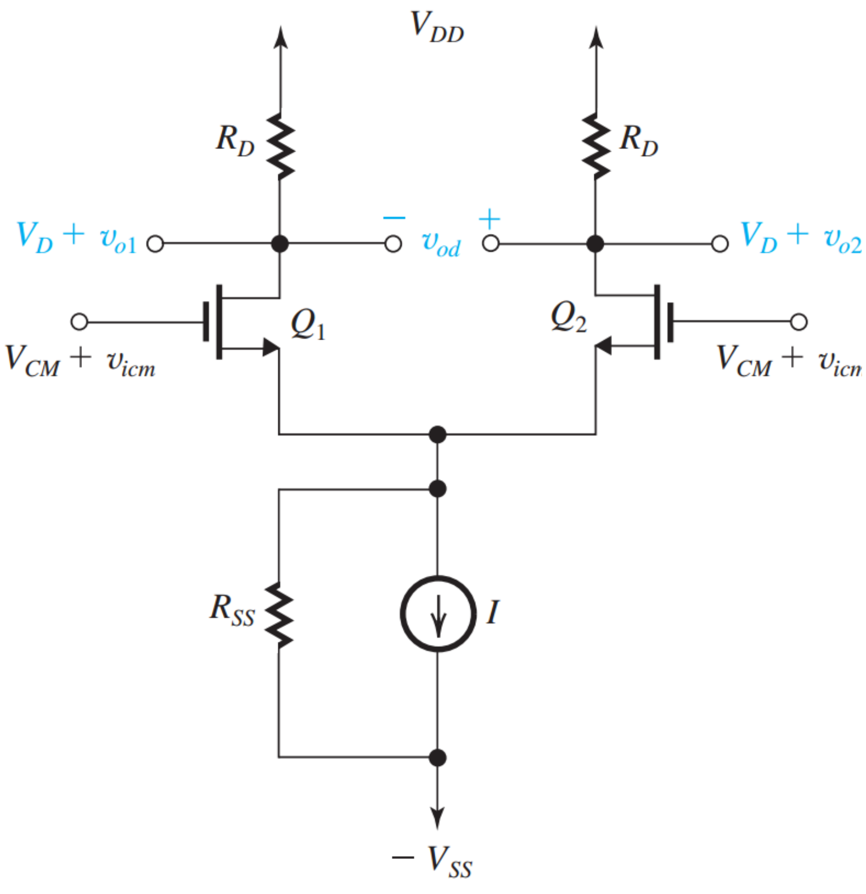
Research Project-deadline June 20th 11:00pm 2026

- Open topic: Choose **ONE** of the following topics for your research project:
- **1. Wireless neural stimulators**
- **2. Ultra-low voltage DC-DC converter for energy harvesting applications**
- **3. Ultra-low power operational amplifiers**
- Step 1: Literature search and review
- Step 2: Analyze the existing methods using provided data in the literatures
- Step 3: Propose something new (optional).
- Step 4: Write a research report (at least 5 pages) using the IEEE format

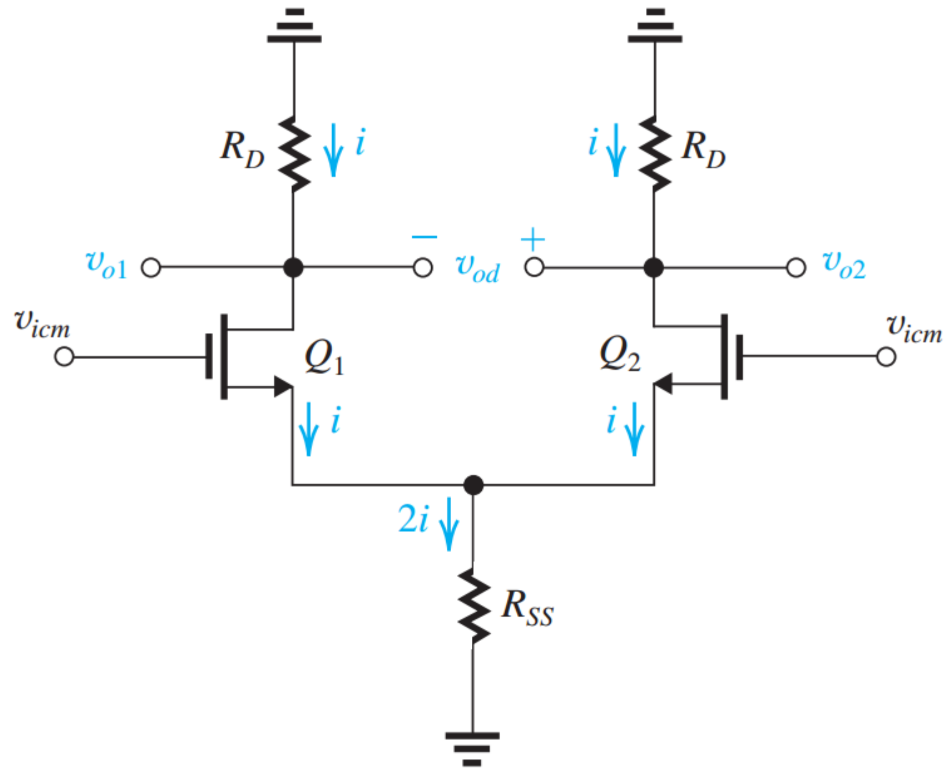
Outline

- Differential amplifiers 2
- SEDTRA/SMITH book page 627-663；

AC Equivalent Circuit for Common Mode Input



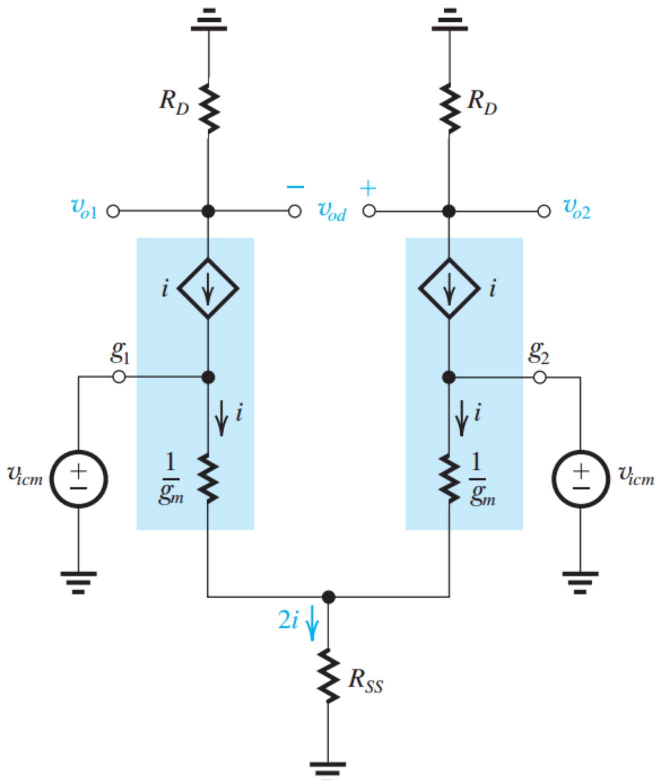
(a)



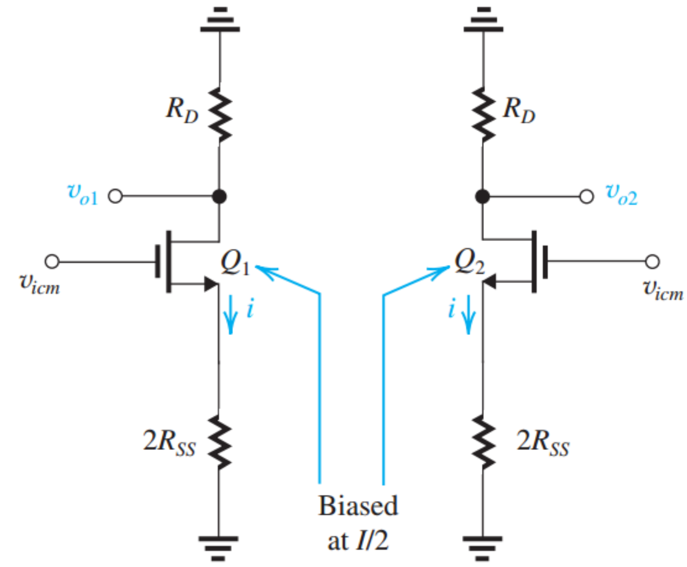
(b)

- Non-ideal current source consists of an ideal current source, shunted by a large resistance, R_{ss}.

Common Mode "Half Circuit"



(c)

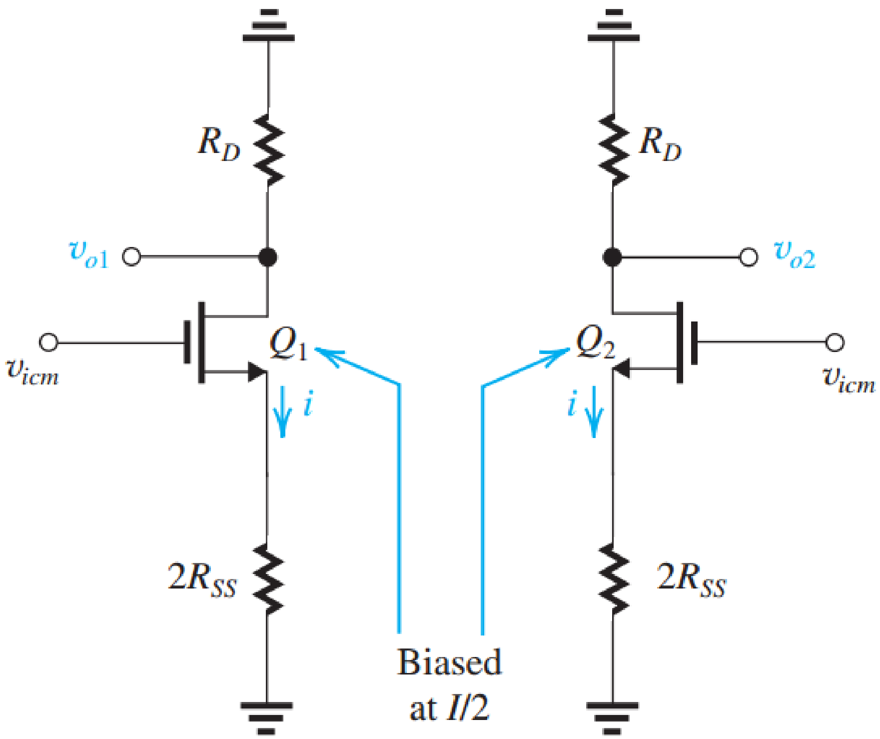


(d)

- For differential inputs, the two half circuits are anti-symmetric, and the joint (Source) is always at virtual ground

- For common-mode inputs, the two half circuits are symmetric. The Source is not virtual ground any more.
- R_{SS} can be considered as two parallel combination of R_{SS}.
- Each CM half circuit has $2R_{SS}$ connected to the source.

Ideal CM Output Voltage



(d)

$$v_{o1} = v_{o2} = -\frac{R_D}{1/g_m + 2R_{SS}} v_{icm}$$

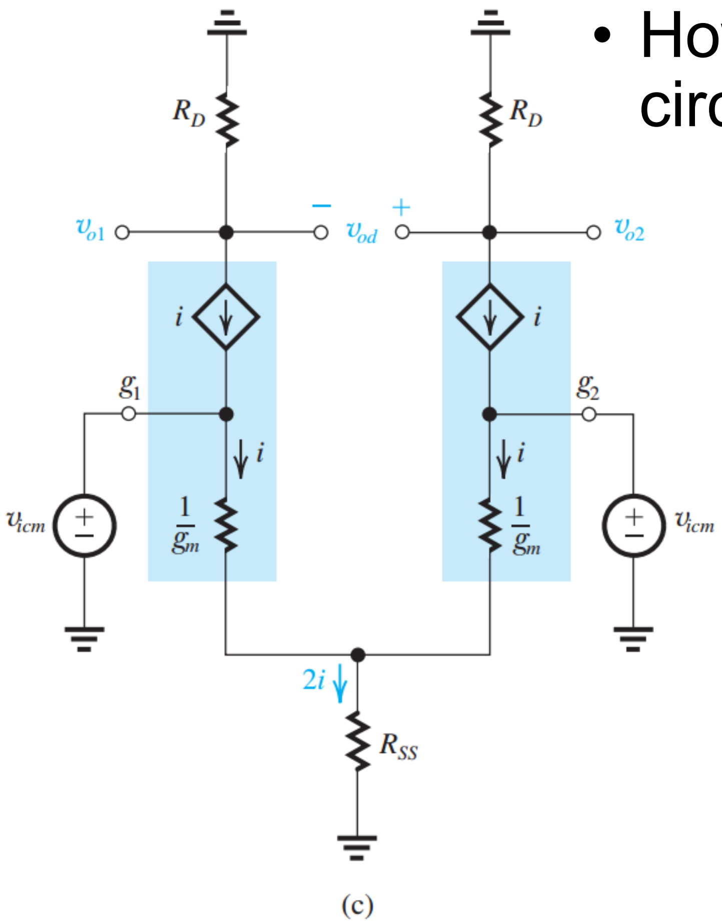
$$\frac{v_{o1}}{v_{icm}} = \frac{v_{o2}}{v_{icm}} \simeq -\frac{R_D}{2R_{SS}}$$

$$v_{od} = v_{o2} - v_{o1} = 0$$

- Common-Source with degeneration

- V_{od} is 0 for ideal diff pair
  - » 1. perfectly matched transistors and resistors
  - » 2. Small CM voltage to keep Q1&Q2 in saturation.

Common Mode Gain with Mismatched RD



- However, any mismatch in the half circuits will produce finite output voltage.

$$v_{o1} \simeq -\frac{R_D}{2R_{SS}} v_{icm}$$

$$v_{o2} \simeq -\frac{R_D + \Delta R_D}{2R_{SS}} v_{icm}$$

$$v_{od} = v_{o2} - v_{o1} = -\frac{\Delta R_D}{2R_{SS}} v_{icm}$$

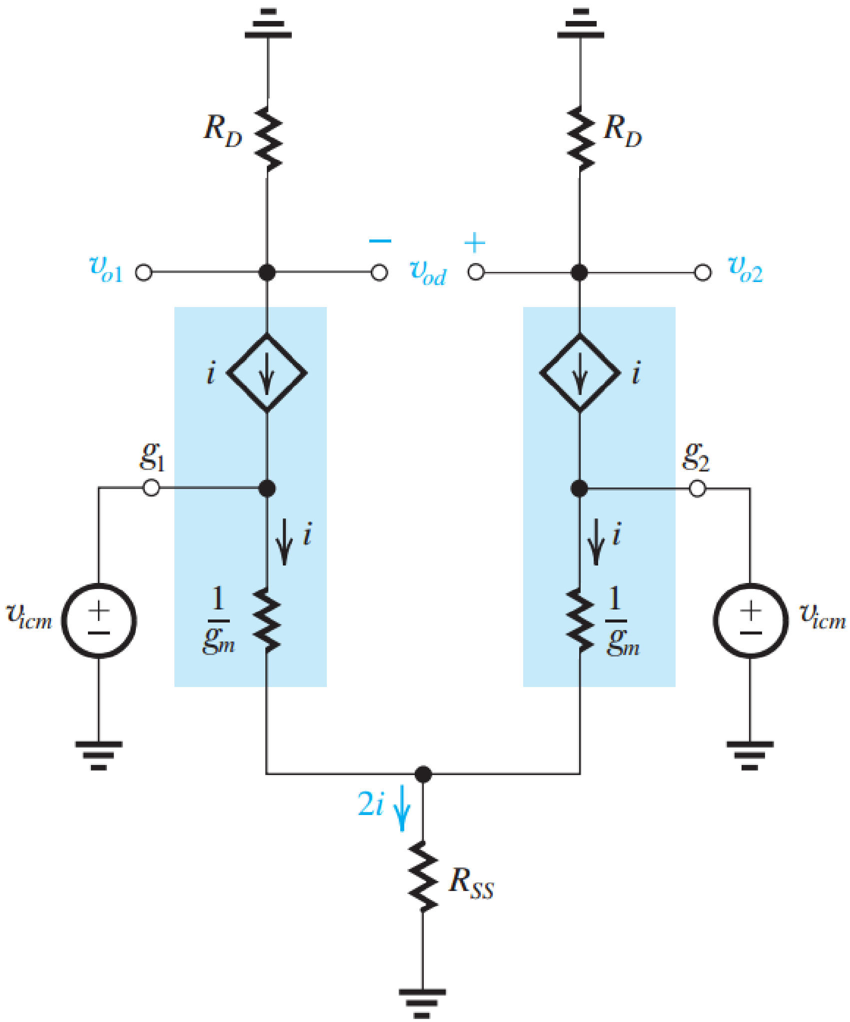
$$A_{cm} \equiv \frac{v_{od}}{v_{icm}} = -\frac{\Delta R_D}{2R_{SS}}$$

$$= -\left(\frac{R_D}{2R_{SS}}\right)\left(\frac{\Delta R_D}{R_D}\right)$$

$$\text{CMRR(dB)} = 20 \log \frac{|A_d|}{|A_{cm}|}$$

$$\text{CMRR} = (2g_m R_{SS}) \Big/ \left(\frac{\Delta R_D}{R_D}\right)$$

Common Mode Gain with Mismatched g_m



(c)

$$g_{m1} = g_m + \frac{1}{2}\Delta g_m$$

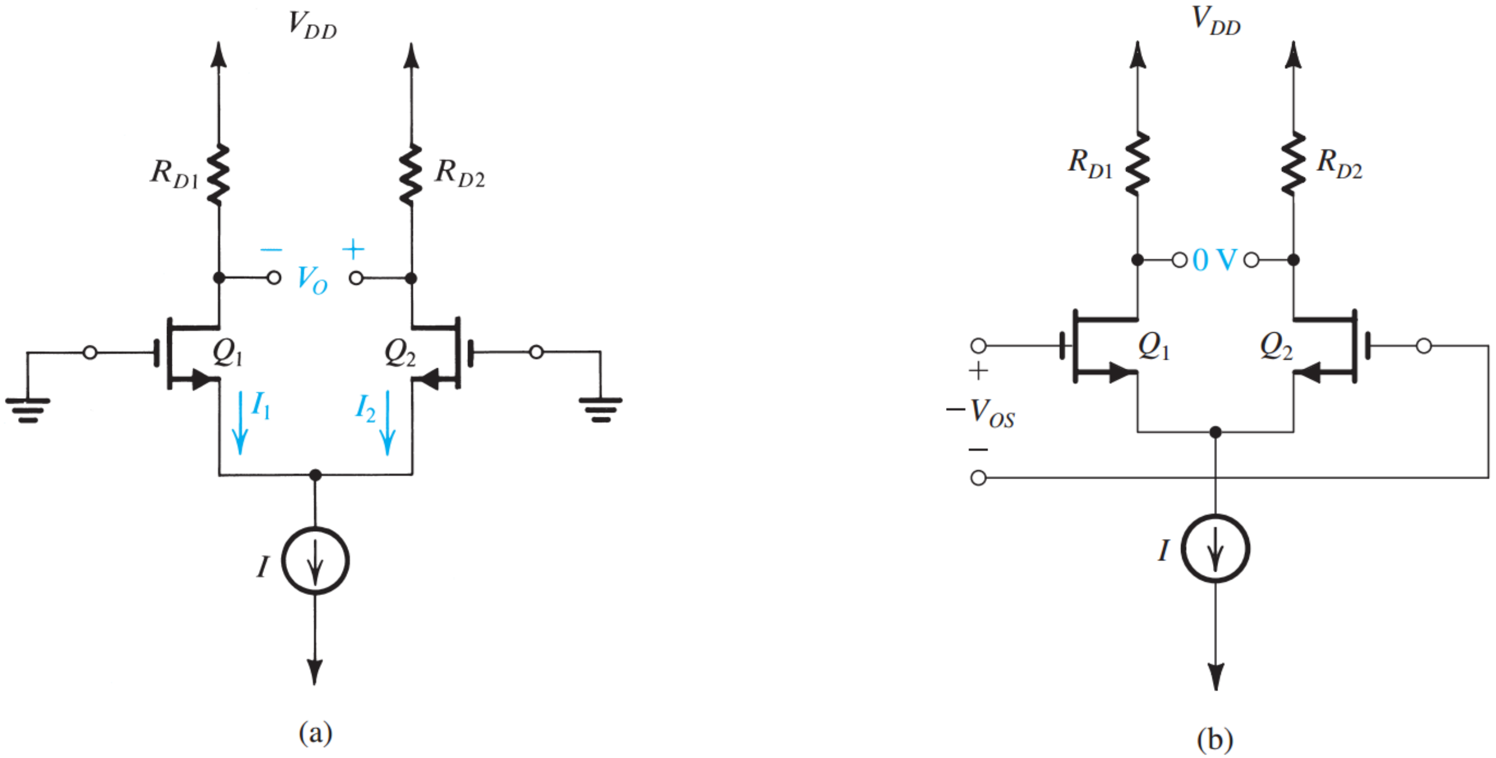
$$g_{m2} = g_m - \frac{1}{2}\Delta g_m$$

$$g_{m1} - g_{m2} = \Delta g_m$$

$$A_{cm} \simeq \left(\frac{R_D}{2R_{SS}}\right)\left(\frac{\Delta g_m}{g_m}\right)$$

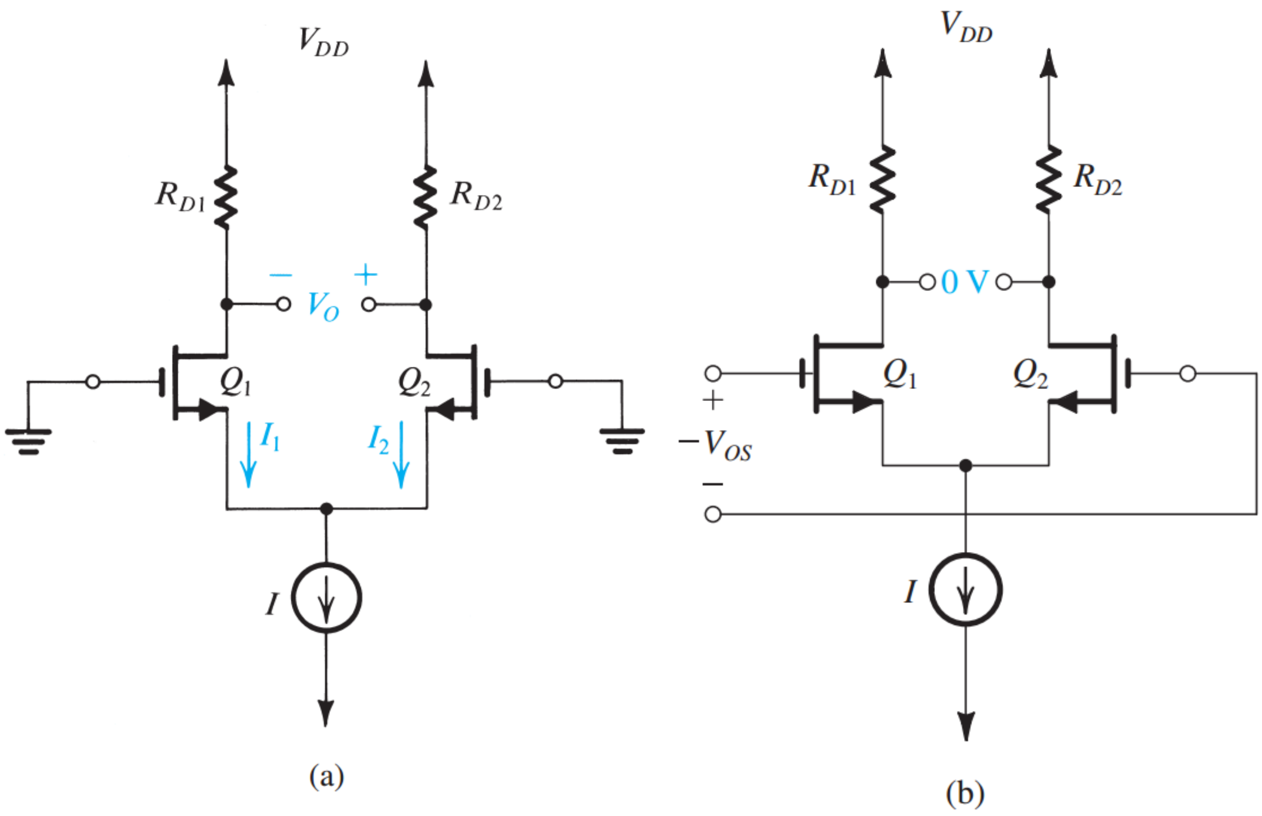
$$\text{CMRR} = (2g_m R_{SS}) \Big/ \left(\frac{\Delta g_m}{g_m}\right)$$

DC Offset



- Due to mismatch in R_D, output voltage $V_o \neq 0$ even when both inputs are grounded.
- To produce 0 output, an input offset voltage $V_{OS} = V_o/A_d$, where A_d is differential gain, needs to be applied.

DC Offset



$$R_{D1} = R_D + \frac{\Delta R_D}{2}$$

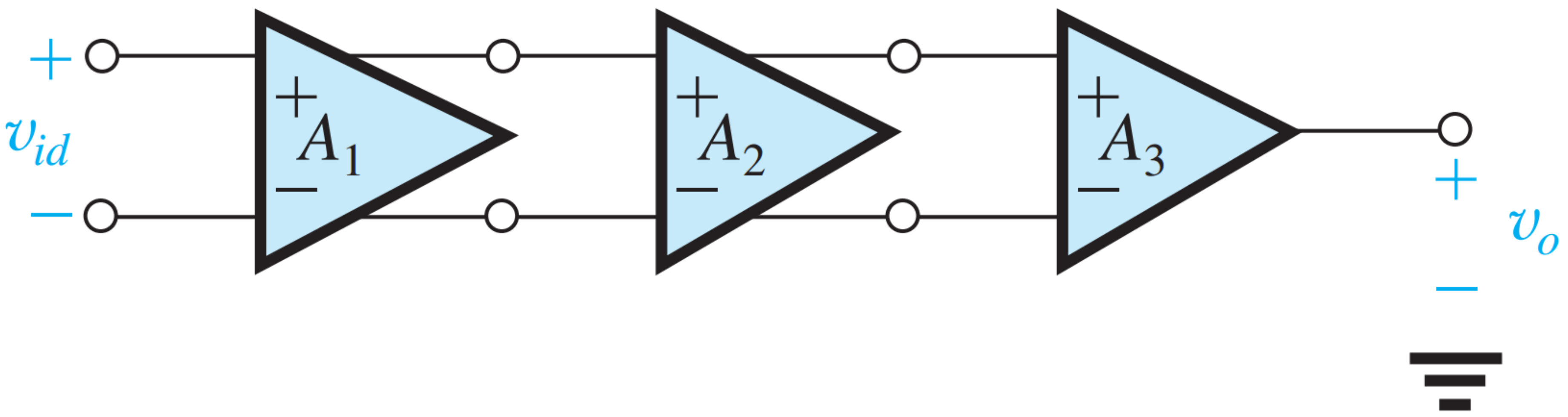
$$R_{D2} = R_D - \frac{\Delta R_D}{2}$$

$$V_O = V_{D2} - V_{D1}$$
$$= \left(\frac{I}{2}\right) \Delta R_D$$

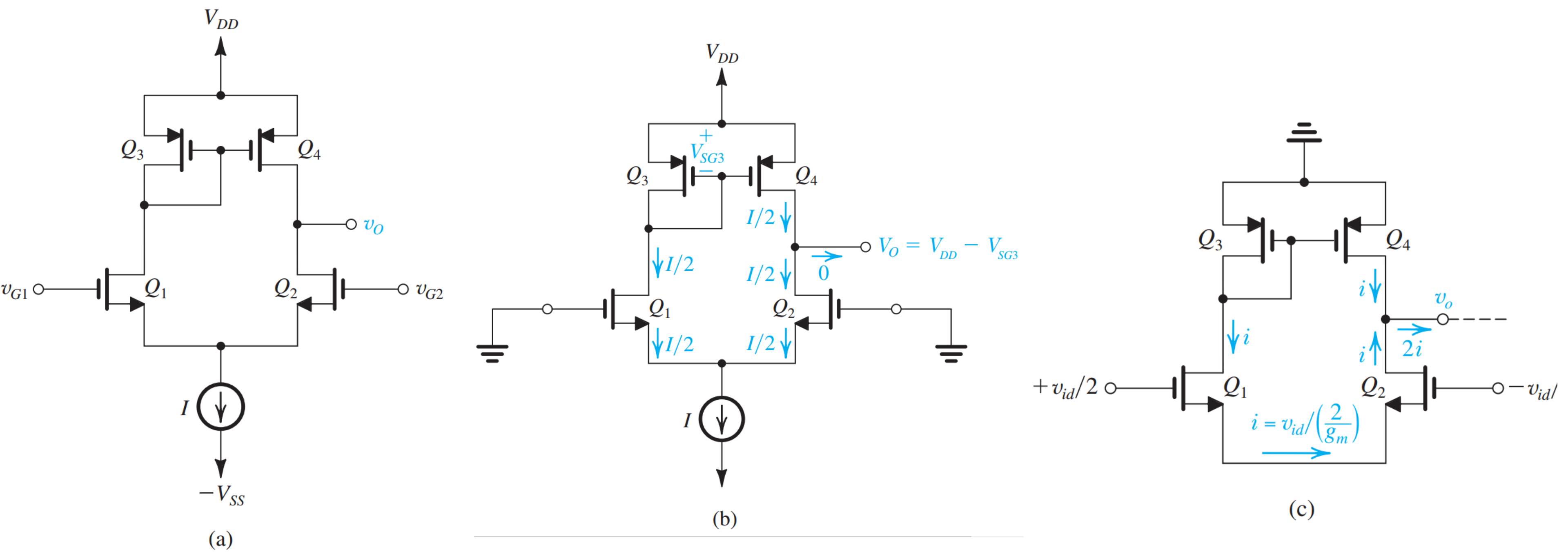
$$V_{OS} = \left(\frac{V_{OV}}{2}\right)\left(\frac{\Delta R_D}{R_D}\right)$$

$$g_m = \frac{2I_D}{V_{OV}} = \frac{2(I/2)}{V_{OV}} = \frac{I}{V_{OV}}$$

Differential Input, Single-End Output

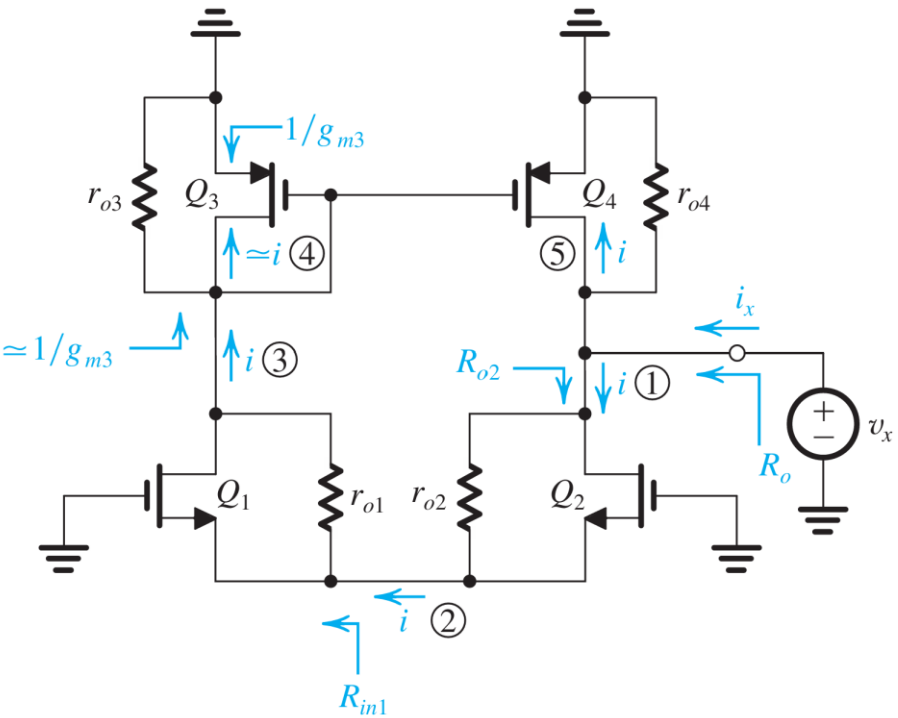


MOS Differential Pair with Current Mirror Load



$$A_d \equiv \frac{v_o}{v_{id}} = G_m R_o = g_{m1,2}(r_{o2} \| r_{o4})$$

MOS Differential Pair with Current Mirror Load



$$R_{in1} = \frac{r_{o1} + R_L}{g_{m1} r_{o1}} = \frac{1}{g_{m1}} + \frac{1/g_{m3}}{g_{m1} r_{o1}} \simeq \frac{1}{g_{m1}}$$

$$R_{o2} = R_{in1} + r_{o2} + g_{m2} r_{o2} R_{in1} = \frac{1}{g_{m1}} + r_{o2} + \left(\frac{g_{m2}}{g_{m1}}\right) r_{o2}$$

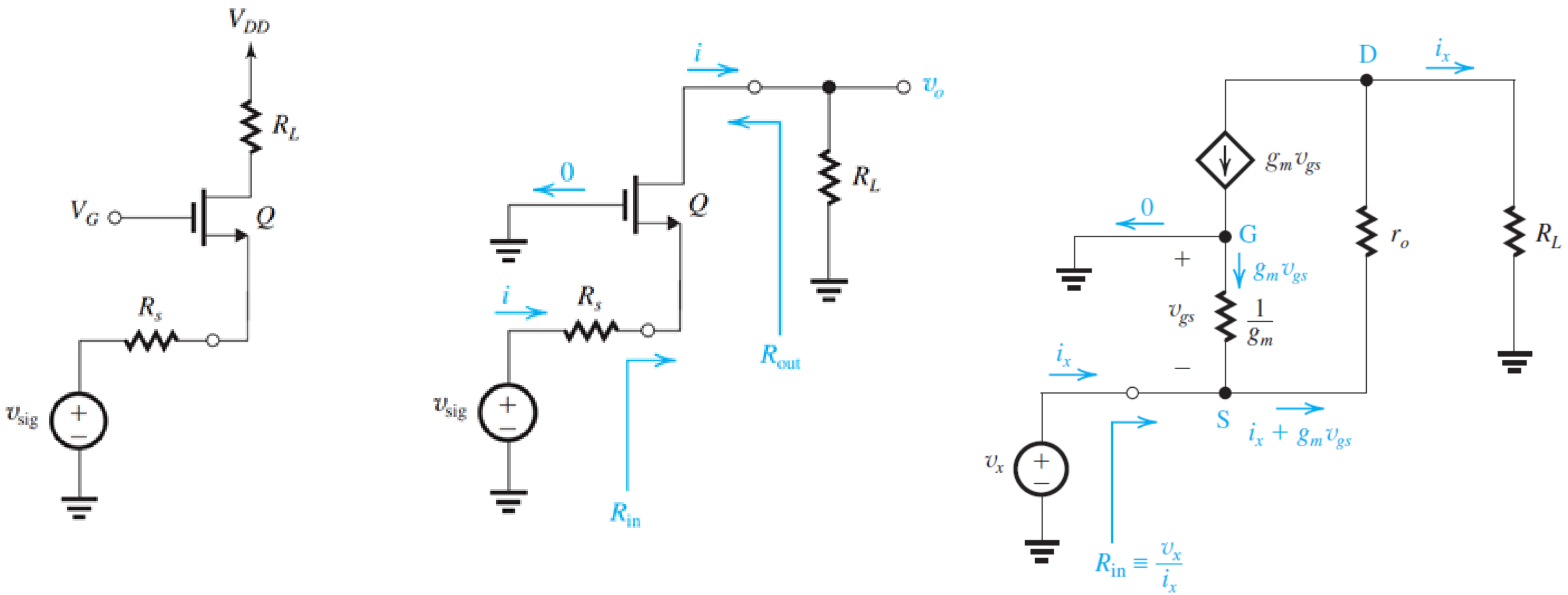
$$R_{o2} \simeq 2 r_{o2}$$

$$i_x = i + i + \frac{v_x}{r_{o4}} = 2i + \frac{v_x}{r_{o4}} = 2\frac{v_x}{R_{o2}} + \frac{v_x}{r_{o4}}$$

$$i_x = 2\frac{v_x}{2r_{o2}} + \frac{v_x}{r_{o4}}$$

$$R_o \equiv \frac{v_x}{i_x} = r_{o2} \parallel r_{o4}$$

Review: Impedance Transformation of Common Gate Amplifier

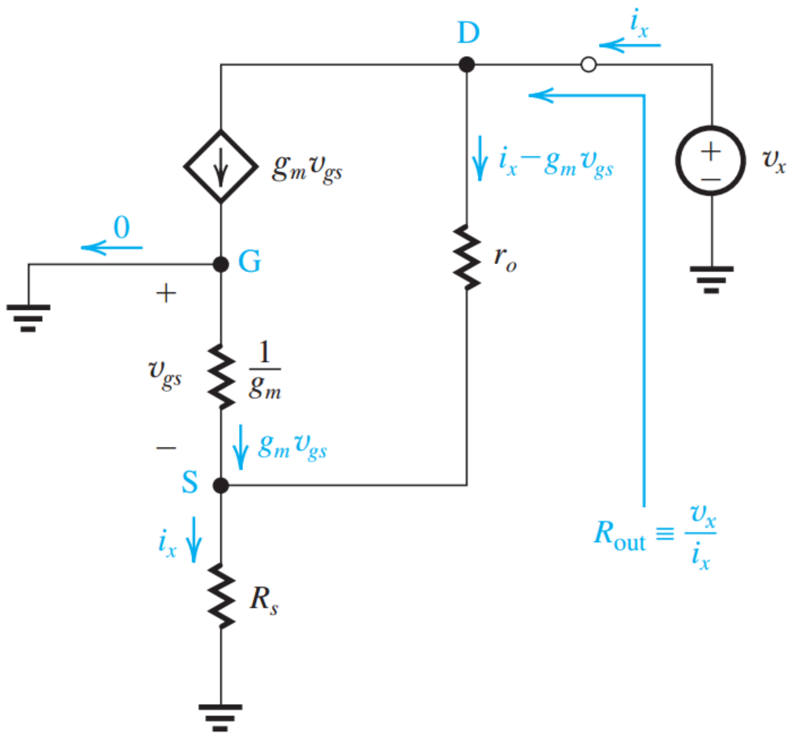


$$v_x = (i_x + g_m v_{gs}) r_o + i_x R_L$$

$$R_{in} = \frac{r_o + R_L}{1 + g_m r_o} \simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

The load impedance is transformed to the input by dividing by the intrinsic gain.

Review: Continued 1



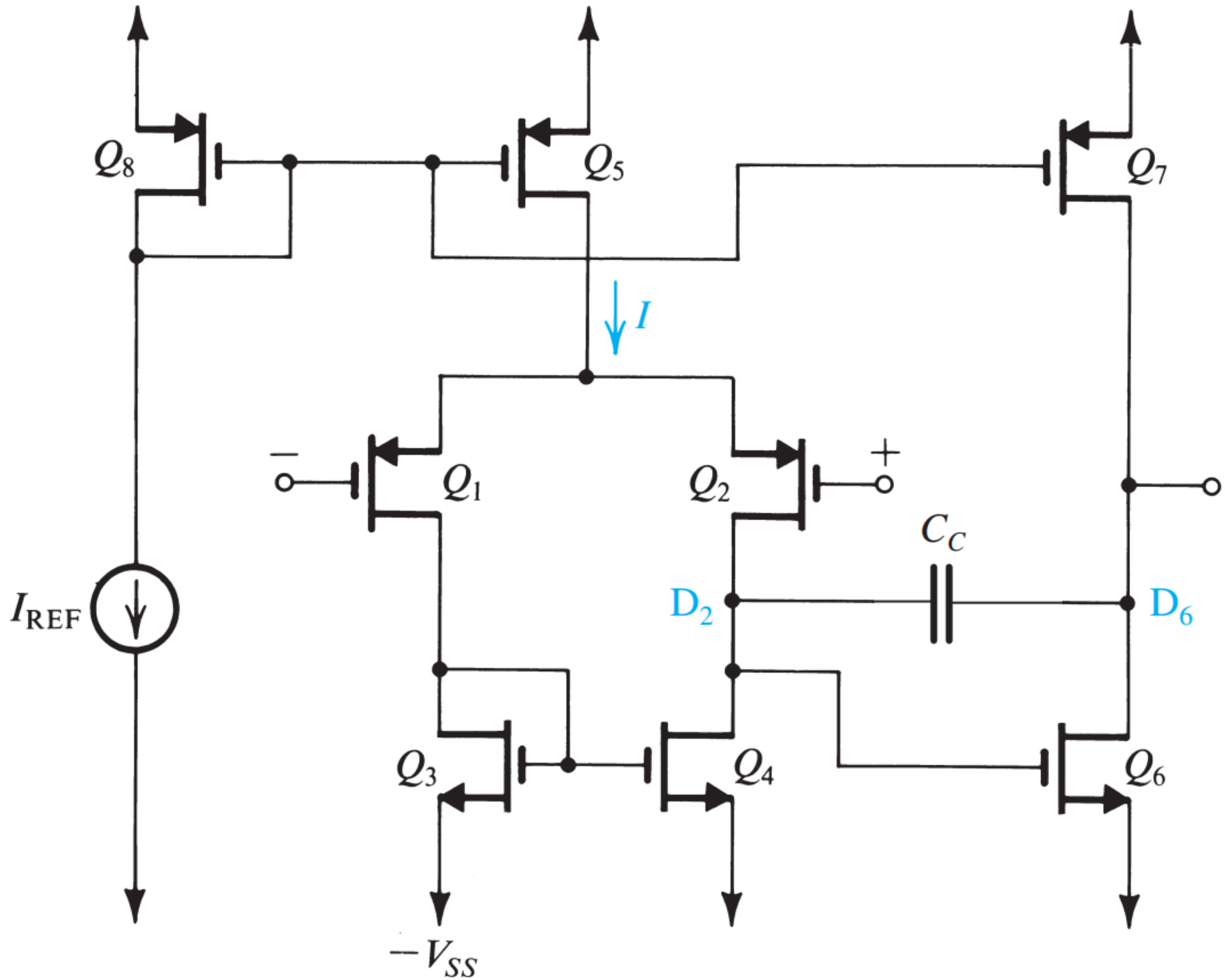
$$v_x = (i_x - g_m v_{gs}) r_o + i_x R_s$$

$$v_{gs} = -i_x R_s$$

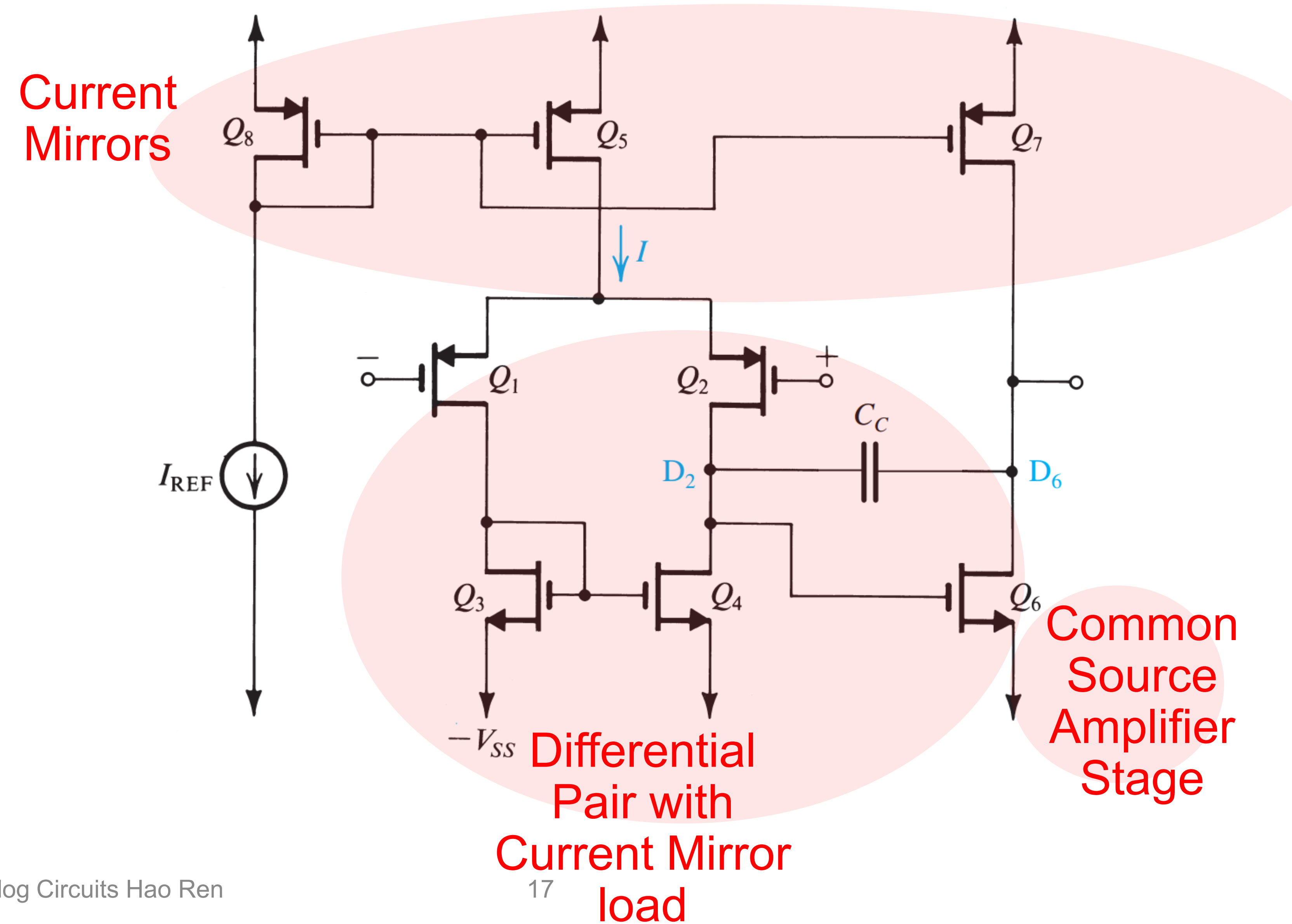
$$\begin{aligned} R_{\text{out}} &= r_o + R_s + g_m r_o R_s \\ &= r_o + (1 + g_m r_o) R_s \\ &\simeq (g_m r_o) R_s \end{aligned}$$

The source impedance is transformed to the output by multiplying the intrinsic gain.

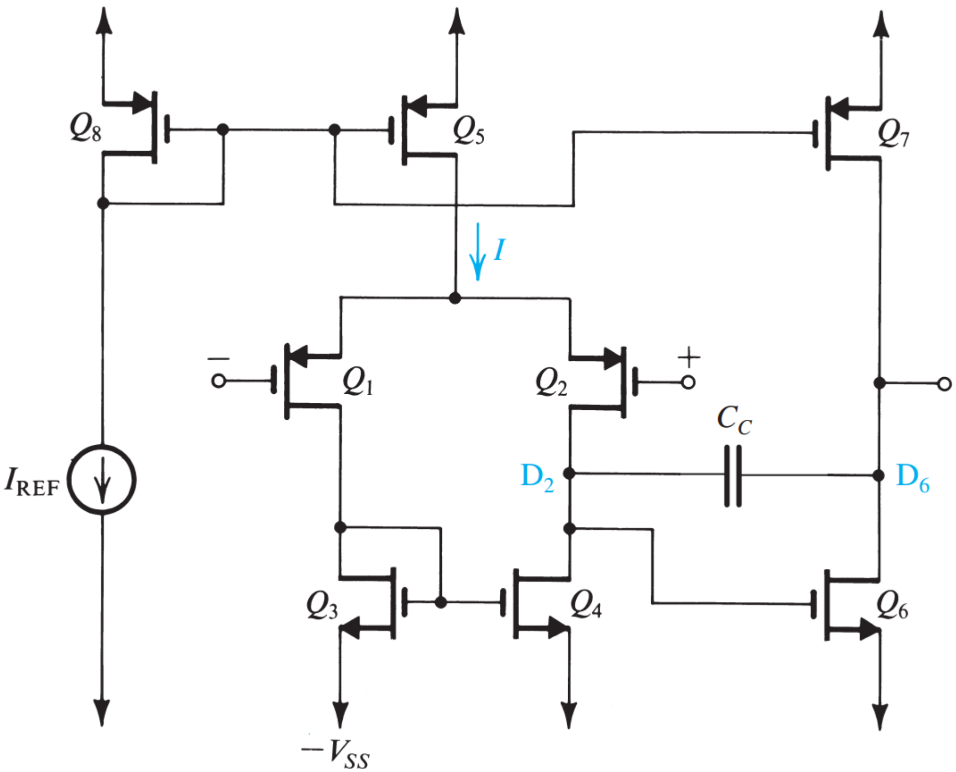
Two-Stage CMOS Op-Amp Circuit



Stages Scrutinize



Voltage Gains



- Voltage gain of the first stage (Q_1, Q_2): Differential input, single-ended output:

$$A_1 = -g_{m1}(r_{o2} \,||\, r_{o4})$$

- Voltage gain of the 2nd stage (Q_6): Common source with current source load:

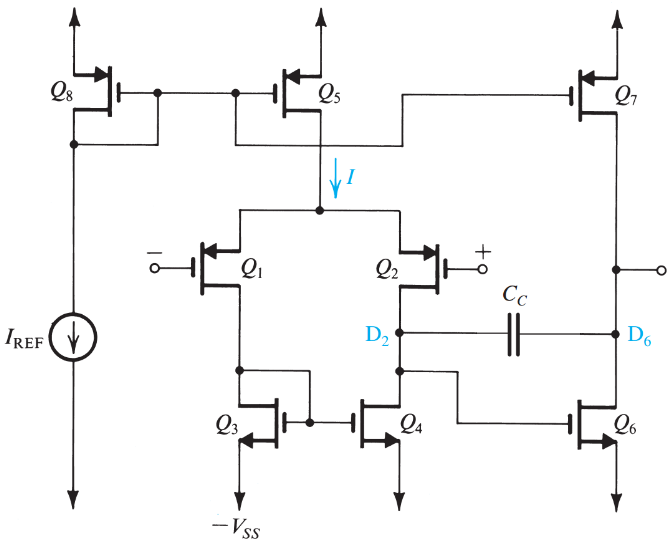
$$A_2 = -g_{m6}(r_{o6} \,||\, r_{o7})$$

- Total gain

$$A_o = A_1 A_2$$

Example:

Transistor	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8
W/L	20/0.8	20/0.8	5/0.8	5/0.8	40/0.8	10/0.8	40/0.8	40/0.8

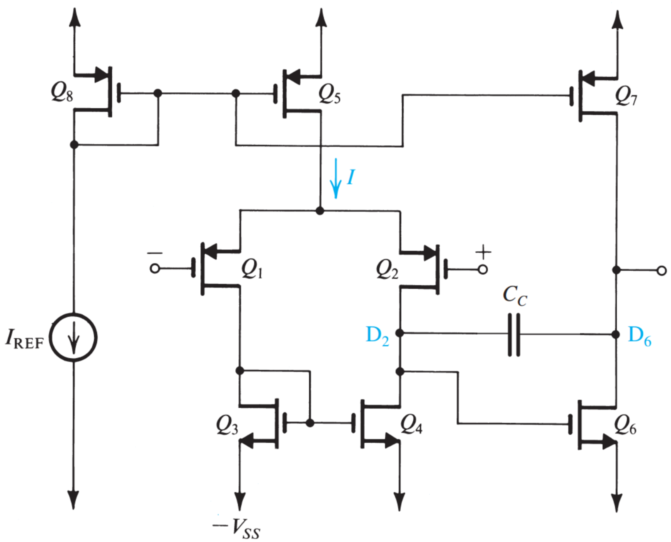


- I_{REF} = 90 μA, V_{tn} = 0.7 V, V_{tp} = -0.8 V, $\mu_n C_{ox}$ = 160 $\mu A/V^2$, $\mu_p C_{ox}$ = 40 $\mu A/V^2$, $|V_A|$ = 10 V for all devices V_{DD} = V_{SS} = 2.5 V.

- Find I_D, $|V_{OV}|$, $|V_{GS}|$, g_m, r_o for all Q's, voltage gain, input common mode range, output voltage range.

Example: solution

Transistor	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8
W/L	20/0.8	20/0.8	5/0.8	5/0.8	40/0.8	10/0.8	40/0.8	40/0.8



Refer to Fig. 9.40. Since Q_8 and Q_5 are matched, $I = I_{REF}$. Thus Q_1, Q_2, Q_3, and Q_4 each conducts a current equal to $I/2 = 45\ \mu A$. Since Q_7 is matched to Q_5 and Q_8, the current in Q_7 is equal to $I_{REF} = 90\ \mu A$. Finally, Q_6 conducts an equal current of $90\ \mu A$.

With I_D of each device known, we use

$$I_D = \frac{1}{2}(\mu C_{ox})(W/L)V_{OV}^2$$

to determine $|V_{OV}|$ for each transistor. Then we find $|V_{GS}|$ from $|V_{GS}| = |V_t| + |V_{OV}|$. The results are given in Table 9.1.

The transconductance of each device is determined from

$$g_m = 2I_D/|V_{OV}|$$

The value of r_o is determined from

$$r_o = |V_A|/I_D$$

The resulting values of g_m and r_o are given in Table 9.1.

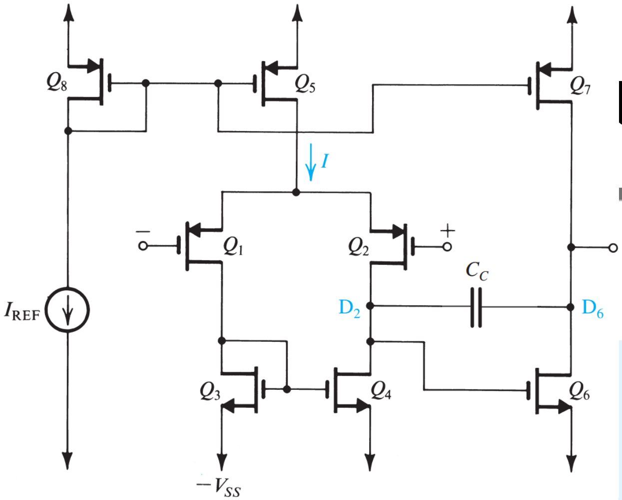


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Table 9.1

	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8
I_D (μA)	45	45	45	45	90	90	90	90
$\lvert V_{OV}\rvert$(V)	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
$\lvert V_{GS}\rvert$(V)	1.1	1.1	1	1	1.1	1	1.1	1.1
g_m (mA/V)	0.3	0.3	0.3	0.3	0.6	0.6	0.6	0.6
r_o (kΩ)	222	222	222	222	111	111	111	111

The voltage gain of the first stage is determined from

$$A_1 = -g_{m1}(r_{o2} \| r_{o4}) \\ = -0.3(222 \| 222) = -33.3 \text{ V/V}$$

The voltage gain of the second stage is determined from

$$A_2 = -g_{m6}(r_{o6} \| r_{o7}) \\ = -0.6(111 \| 111) = -33.3\text{V/V}$$

Thus the overall dc open-loop gain is

$$A_0 = A_1 A_2 = (-33.3) \times (-33.3) = 1109 \text{ V/V}$$

or

$$20 \log 1109 = 61 \text{ dB}$$

The lower limit of the input common-mode range is the value of input voltage at which Q_1 and Q_2 leave the saturation region. This occurs when the input voltage falls below the voltage at the drain of Q_1 by $|V_{tp}|$ volts. Since the drain of Q_1 is at $-2.5 + 1 = -1.5$ V, then the lower limit of the input common-mode range is -2.3 V.

The upper limit of the input common-mode range is the value of input voltage at which Q_5 leaves the saturation region. Since for Q_5 to operate in saturation the voltage across it (i.e., V_{SD5}) should at least be equal to the overdrive voltage at which it is operating (i.e., 0.3 V), the highest voltage permitted at the drain of Q_5 should be +2.2 V. It follows that the highest value of v_{ICM} should be

$$v_{ICM\max} = 2.2 - 1.1 = 1.1\text{V}$$

The highest allowable output voltage is the value at which Q_7 leaves the saturation region, which is $V_{DD} - |V_{OV7}| = 2.5 - 0.3 = 2.2$ V. The lowest allowable output voltage is the value at which Q_6 leaves saturation, which is $-V_{SS} + V_{OV6} = -2.5 + 0.3 = -2.2$ V. Thus, the output voltage range is -2.2 V to $+2.2$ V.

Homework7-due May 27th 11p.m.

SEDRA/SMITH Book chapter 9 problem 9.2, 9.6, 9.19, 9.59, 9.60, 9.71,